SQL **Text-to-SQL Agent**

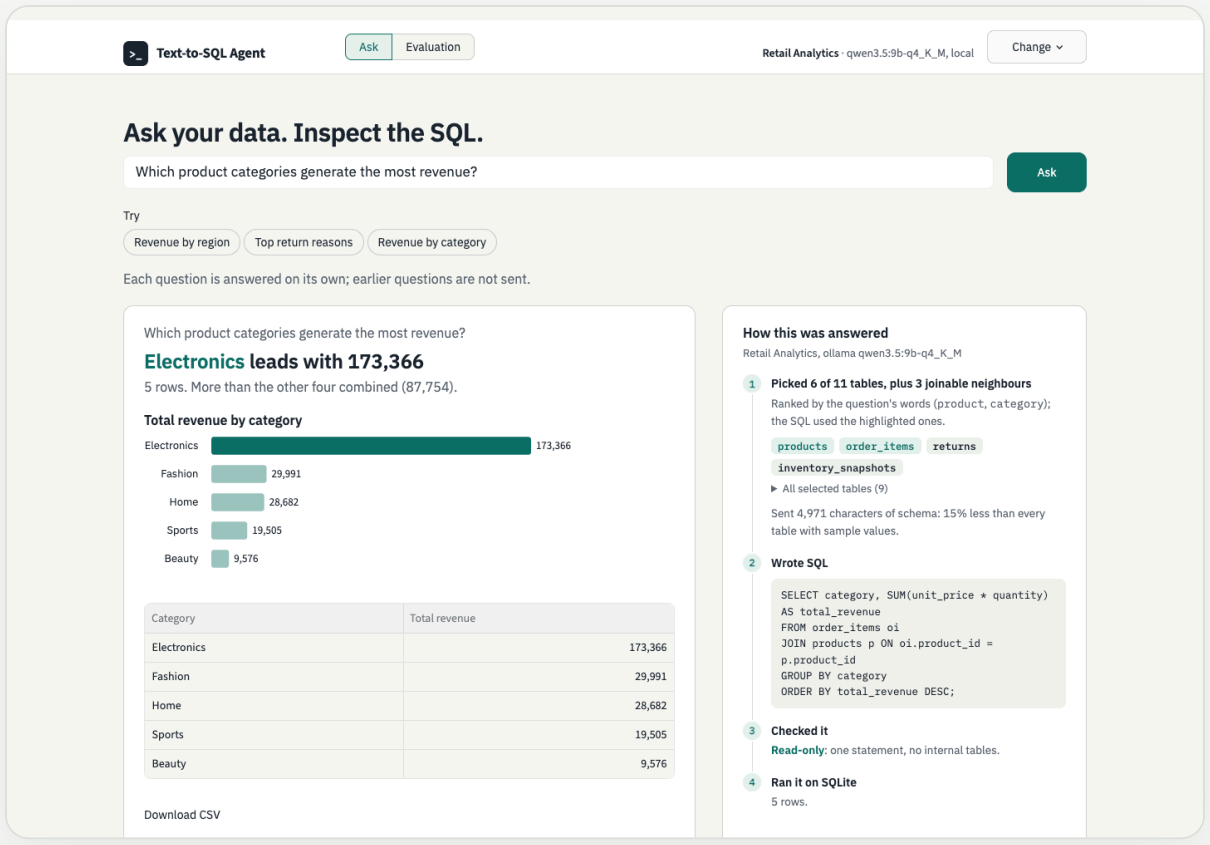
Portfolio prototype · results as of 5 Nov 2026

ASK IN ENGLISH, CHECK THE SQL

Answers you can inspect, from a database it can only read.

A local or hosted model writes one SELECT; Python decides whether it runs. Every answer shows the rows, the SQL, and the exact schema text the model was given. Result rows are never sent to the model.

SQLite · DuckDB · PostgreSQL · Ollama or Gemini



HOW AN ANSWER IS PRODUCED

01 Read schema
The engine builds one chunk per table: DDL, keys, and up to six frequent values of low-cardinality text columns.

02 Pick tables
BM25, name matches and character trigrams rank tables; the top 6 plus foreign-key neighbours are sent.

03 Generate
The model returns one SQL statement for that schema and the question, nothing else.

04 Read-only gate
AST validation in the engine's dialect, then the engine's own read-only mode, a work budget and a row cap.

05 Show the work
Rows, a chart where it fits, the SQL, the tables picked and why, and one repair try on a failed run.

MEASURED, WITH 95 % INTERVALS

Standard benchmarks, full dev sets

Local `qwen3.5:9b` (Ollama, thinking off), typed result comparison. Not leaderboard-comparable.

64.6%	35.2%	25.1%
Spider 1.0 dev 1,034 questions	BIRD dev with evidence 1,534 questions	BIRD dev without evidence 1,534 questions

6 / 6 data-change requests refused

45 / 0 engine conformance tests passed / skipped, live PostgreSQL 16

Large schemas: where choosing tables pays

`bird_train_large`: 1,222 BIRD train questions, 4 databases of 21 to 65 tables.

Full DDL plus value hints	25.5%
Plain full DDL	26.8%
Top 6 tables (default)	29.1%

+3.6 pts [+1.4, +5.8] vs full DDL plus hints, paired

69 % fewer prompt tokens

0.891 retrieval MRR after the scorer update

WHAT WAS TRIED SINCE OCTOBER, EACH PAIRED AGAINST A NAMED BASELINE

CHANGE	EFFECT ON THE 1,222 QUESTIONS	DECISION
Question value matching (reads stored values)	+0.4 pts [-0.9, +1.6] vs top 6	off
Dense embeddings in the table scorer	Questions with a needed table not sent 128 → 74; answers +0.2 pts [-1.6, +2.0] vs the kept scorer	off
Shared prompt layout (every table, the selected ones named)	Latency 10.2 s → 4.3 s; answers −2.4 pts [−4.6, −0.2] vs a fresh top-6 run	opt-in
Shared layout plus each table's columns and joins	−2.1 pts [−4.3, +0.0] vs that top-6 run; across commits, not controlled	opt-in

WHAT THE NUMBERS SAY

Selection pays only at scale.
On Spider and BIRD dev (at most 14 tables) choosing tables added nothing measurable; on 34- and 65-table databases it is the one measured accuracy gain.

Better retrieval is not better answers.
Embeddings sent the right tables far more often, but flipped as many other answers the wrong way. Recall is reported separately from accuracy.

Speed and accuracy trade.
Sending the same schema every time lets Ollama reuse its cache and is 58 % faster, but costs accuracy, so the default stays per-question selection.

Live demo aipa-text-to-sql-agent.streamlit.app
Code, runs and decision log: github.com/tuannm3812/text-to-sql-agent

Python · Streamlit · sqlglot · Ollama / Gemini
Every figure links to a committed run with its manifest